

Module13 - Standard Costing

Standard costs are budgeted cost per individual cost item. A budget is built up of many standard costs

Standard costs are compared to actual costs and explanations are sought for any differences between the two

Variance = difference between budgeted (or standard) cost and actual cost

Adverse variance is where actual is above standard
Favourable variance is where actual is below standard cost

Variances occur in:

- 1). Material & Labour
- 2). Variable & fixed overheads
- 3). Sales

1) The anatomy of variances: materials & labour

1) Material variances

Material cost can be more or less than standard for 2 reasons:

- 1) actual production used more or less material than planned
- 2) The price to purchase the material was more or less than planned.

(SQ-AQ)xSP
(Standard quantity - actual quantity) x standard price per unit price

Reason for using SP after brackets - by holding all prices as standard mngt can gain insight into the efficiency of production

an adverse material efficiency variance may be caused by the use of inferior quality materials which would be reflected in a favourable material price variance

(SP-AP)xAQ
(Standard Price per unit - actual Price per unit) x Actual quantity used

Reason for using AQ after brackets - SQ would not be realistic or actual status

AQ - for production mng use actually produced - for purchasing mng use actually purchased

To address reason # 1

Material Efficiency Variance

To address reason # 2

Material Price Variance

2) Labour variances

Labour cost can be more or less than standard for 2 reasons:

- 1) actual production used more or less time than planned
- 2) The labour rates actually paid were more or less than planned.

(ST-AT)xSR
(Standard Time allowed - actual Time taken) x standard rate per hour

a favourable labour rate variance may be caused by employing less skills operatives

(SR-AR)xAT
(Standard Rate per hour - actual Rate per hour) x Actual Time taken

To address reason # 1

Labour Efficiency Variance

To address reason # 2

Labour Rate Variance

3). Sales Variances

Contribution Variance =

Difference in contribution margin per unit x actual sales in units

Volume variance =

(actual sales - budgeted sales) x budgeted contribution margin per unit

OR Sales quantity variance + sales mix variance

Sales Quantity variance =

(Actual sales - budgeted sales) x budgeted weighted average contribution margin per unit

Sales mix variance =

(Actual sales - budgeted sales) x budgeted contribution margin per unit - budgeted weighted average contribution margin per unit

2) Variable and Fixed overhead analysis

Moves up and down in volume in proportion to productive output as measured by the allocation key used. (e.g direct LH, Direct LC or Machine Hours)

Typically electricity, indirect labour, repairs, maintenance, consumables & suppliers

Variable Overheads variance

Variable Overhead Efficiency Variance

Where: Standard Cost of flexible budget allowance for units produced = Unit produced x allocation key x Std variance rate

Where: Standard Cost of actual time taken for units produced = actual time take (allocation key) x std variance rate

therefore: (Units x Allocation key x Std variance rate) - (actual time taken (allocation key) x std variance rate)

(Standard cost of actual time taken for units produced - Actual variable costs incurred)

Where: Standard Cost of actual time taken for units produced = Actual time taken (allocation key) x std variance rate

Therefore: (Actual time taken x Std variance rate) - (actual variable costs incurred)

Fixed overheads for the year / allocation key (e.g machine hours , DLC or DLH)

Fixed rate per hour

Spending = Budgeted amount - actual amount

Denominator = budgeted amount - Amount applied to units produced

Where: Amount applied to units produced = Actual produced x allocation key

Fixed Overheads Variance

Typically depreciation of plant & equipment, supervisory salaries, factory space costs

Do not vary with production levels within range of output

A favourable denominator variance in fixed overheads may provoke the production staff to aim for even higher levels

Fixed Overhead spending